

Full Stack Development with MERN

Project Documentation

1. Introduction

Project Title: I-Movies : Movie Ticket Booking System

Team Members:

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2. Project Overview

Purpose:

- The primary objective of the I-Movies: Movie Ticket Booking System is to provide users with a seamless and efficient platform for browsing movies, selecting showtimes, and booking tickets from the convenience of their homes. By leveraging the MERN (MongoDB, Express.js, React.js, Node.js) stack, the application delivers a full-featured, responsive, and interactive experience. The goal is to replicate and enhance the traditional ticket booking process, making it faster, user-friendly, and more accessible, especially for users with busy schedules.
- This project was developed as a full stack web application to demonstrate proficiency in modern web technologies, including RESTful API development, secure user authentication, and responsive front-end design.

Features:

- User Authentication & Authorization
- Movie Browsing
- Showtime & Location Selection
- Interactive Seat Selection

3. Architecture

Frontend: React.js SPA with reusable components, Axios for API integration, and React Router for navigation.

Backend: Node.js with Express.js to handle RESTful APIs, route management, and middleware integration.

Database: MongoDB with Mongoose for schema modeling and interactions. Firebase used for authentication and image storage.

System Flow:

User ↔ React.js (Client) ↔ Express.js (Server) ↔ MongoDB

Firestore (Auth + Storage)

4. Setup Instructions

Prerequisites:

- Node.js and npm
- MongoDB
- Firebase project for Authentication and Storage

Installation:

1. Clone the repository.
 2. Run `npm install` in both `/client` and `/server` directories.
 3. Create a `.env` file with necessary environment variables (Mongo URI, Firebase credentials).
 4. Start the backend: `npm run server`
 5. Start the frontend: `npm start`
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5. Folder Structure

Client (React Frontend):

- `/components` – Reusable UI components
- `/pages` – Page-level components (e.g., Home, Movies, Booking)
- `/services` – Axios services for API communication

Server (Node.js Backend):

- `/routes` – API endpoint definitions
 - `/models` – Mongoose schemas for MongoDB
 - `/controllers` – Logic for API routes
-

6. Running the Application

Frontend:

```
cd client
```

```
npm start
```

Backend:

```
cd server
```

```
npm start
```

7. API Documentation

Auth Routes:

- POST /api/register – Register user
- POST /api/login – Login user

Movie Routes:

- GET /api/movies – List movies
- POST /api/movies – Add new movie

Show Routes:

- GET /api/shows – List all shows
- POST /api/shows – Create a new show

Booking Routes:

- POST /api/bookings – Create a booking
- GET /api/bookings/:userEmail – Get bookings for a user

Favorite Routes:

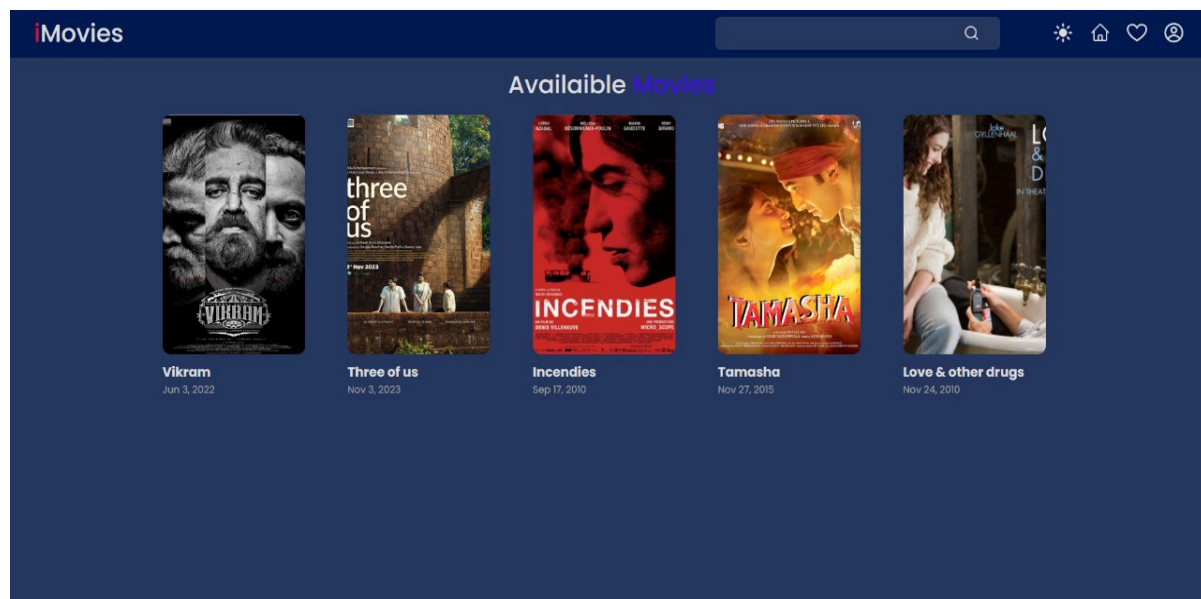
- POST /api/favorites – Add favorite movie
 - GET /api/favorites/:userEmail – Get favorite movies
-

8. Authentication

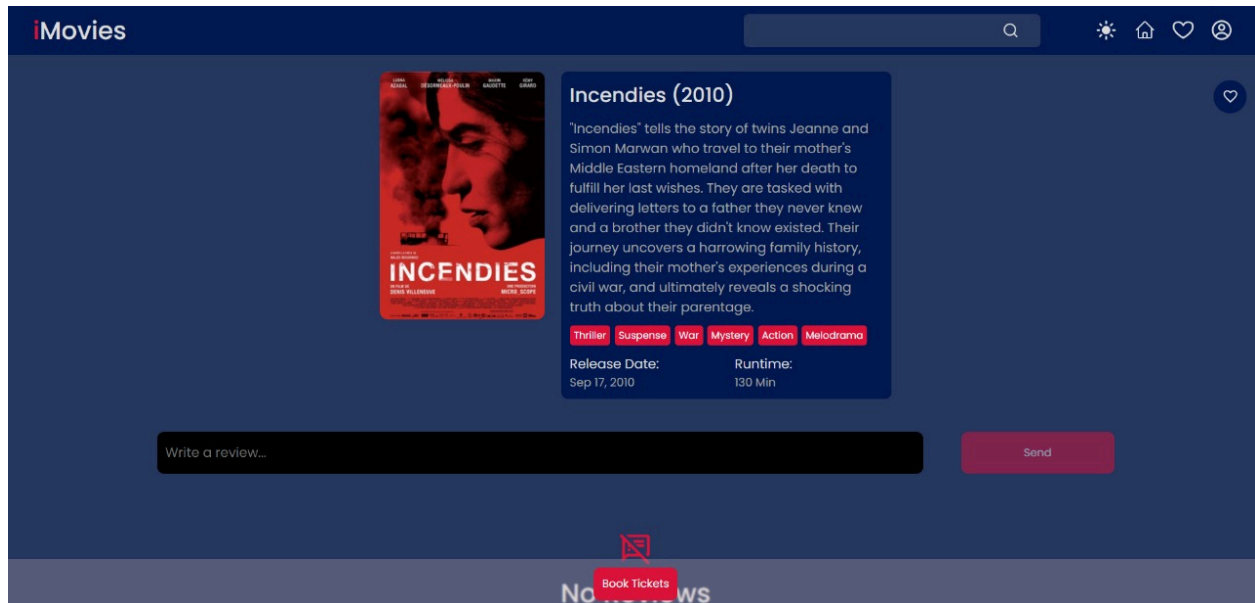
- Used Firebase Authentication for secure login/signup.
 - Authentication token stored client-side.
 - Firebase rules control access to storage.
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9. User Interface

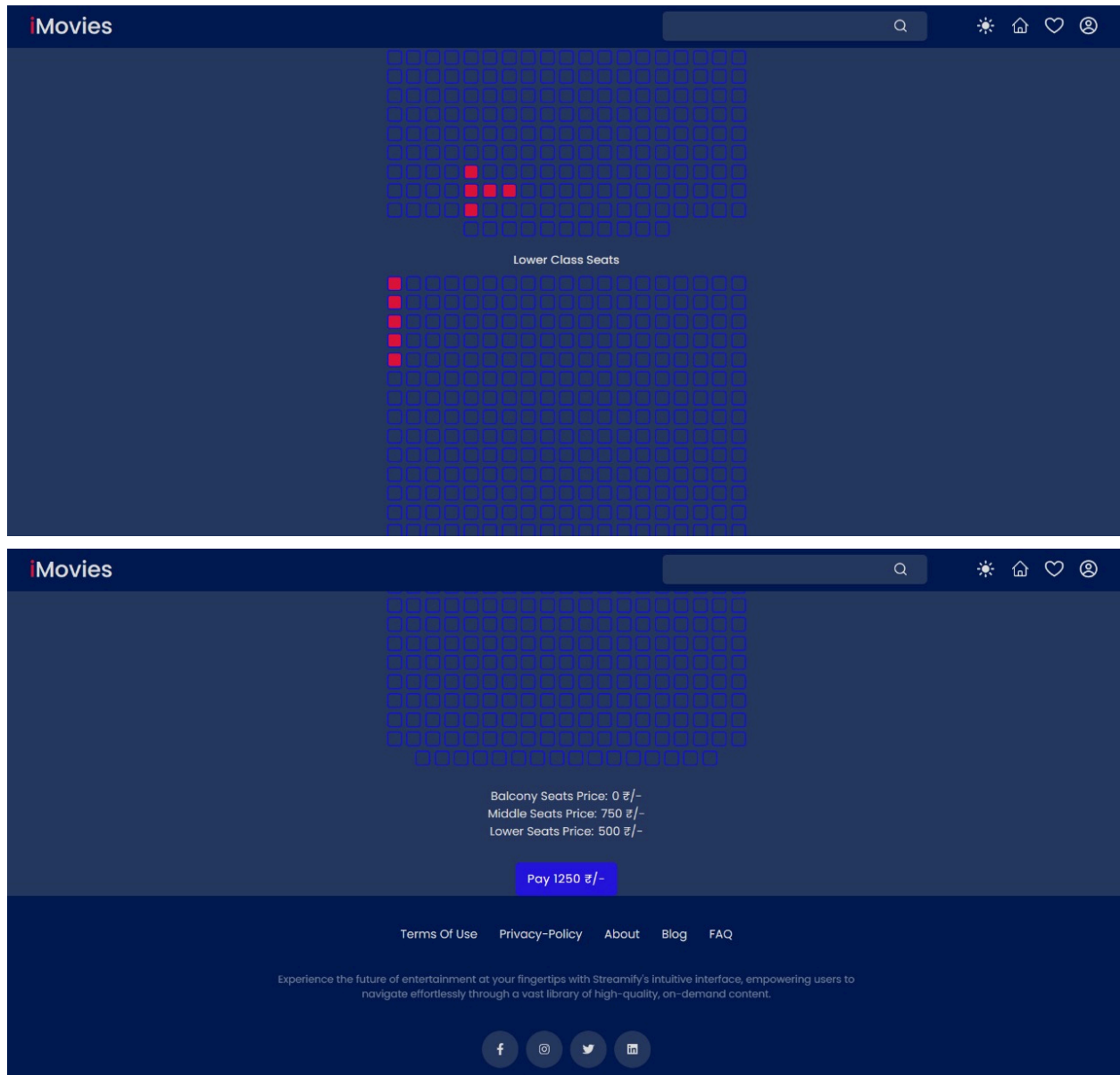
- Home Page: Featured movies, navigation bar



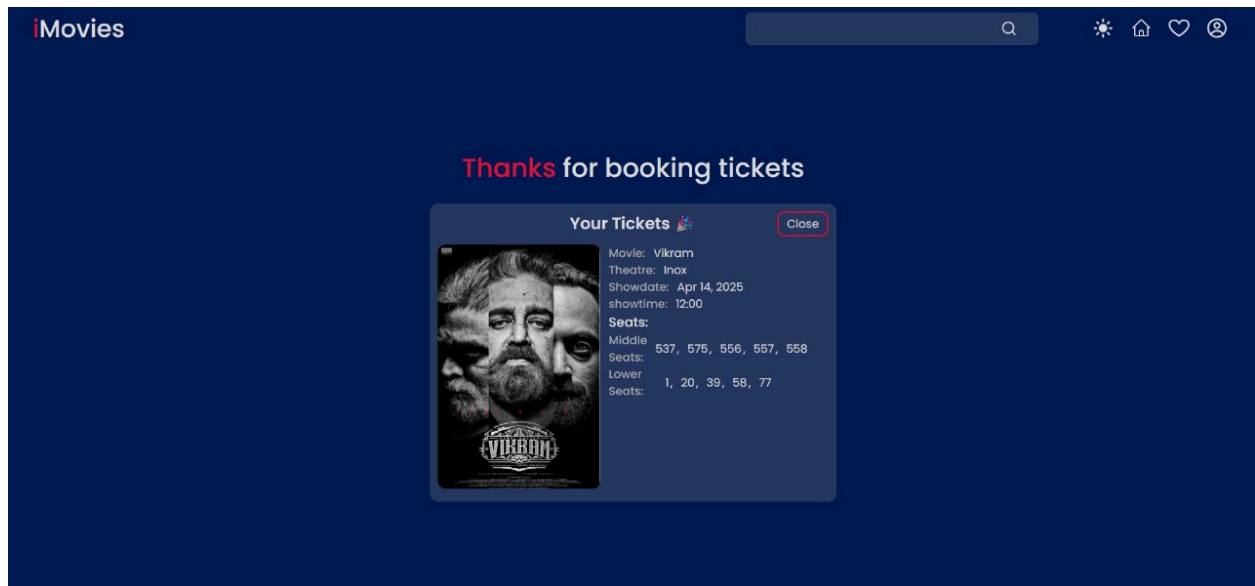
- Movie Detail Page: Trailer, genres, description, showtimes



- Seat Selection: Real-time availability and interactivity



- Booking Confirmation: Displays booking ID and ticket details



10. Testing

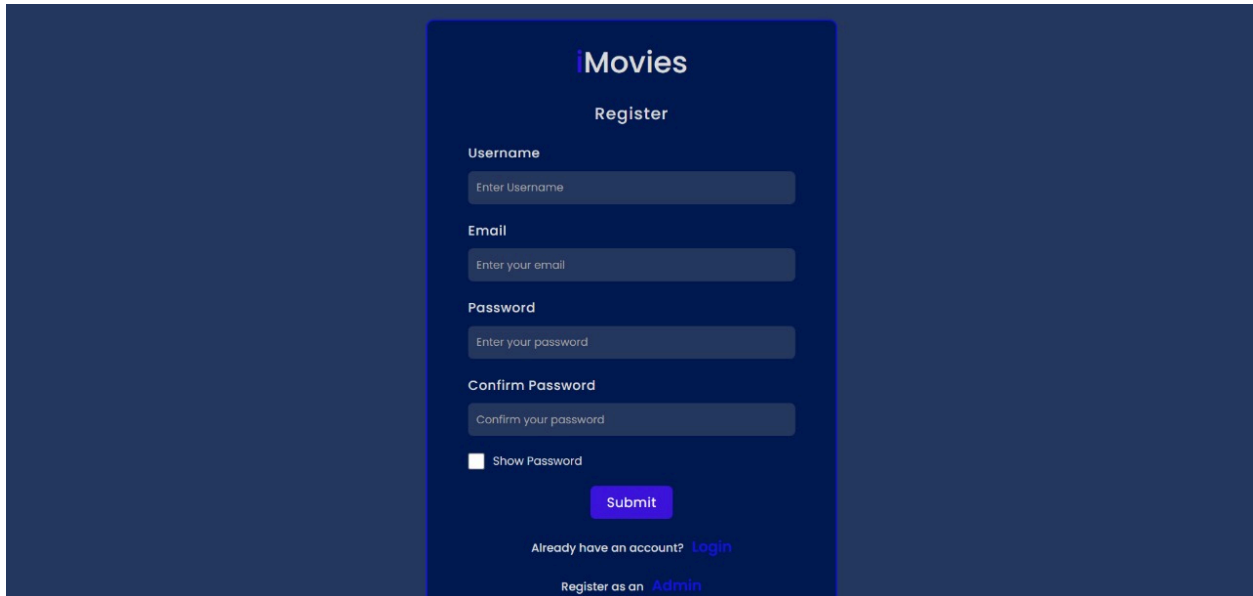
Manual Testing:

- Functionality verified by simulating user journeys.

Postman:

- Used for testing API endpoints during backend development.
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11. Screenshots or Demo



The screenshot shows a registration form for 'iMovies' on a dark blue background. The form is centered and contains the following elements: a title 'iMovies' with a logo, a subtitle 'Register', four input fields for 'Username', 'Email', 'Password', and 'Confirm Password', a 'Show Password' checkbox, a 'Submit' button, a 'Login' link, and an 'Admin' link.

iMovies

Register

Username
Enter Username

Email
Enter your email

Password
Enter your password

Confirm Password
Confirm your password

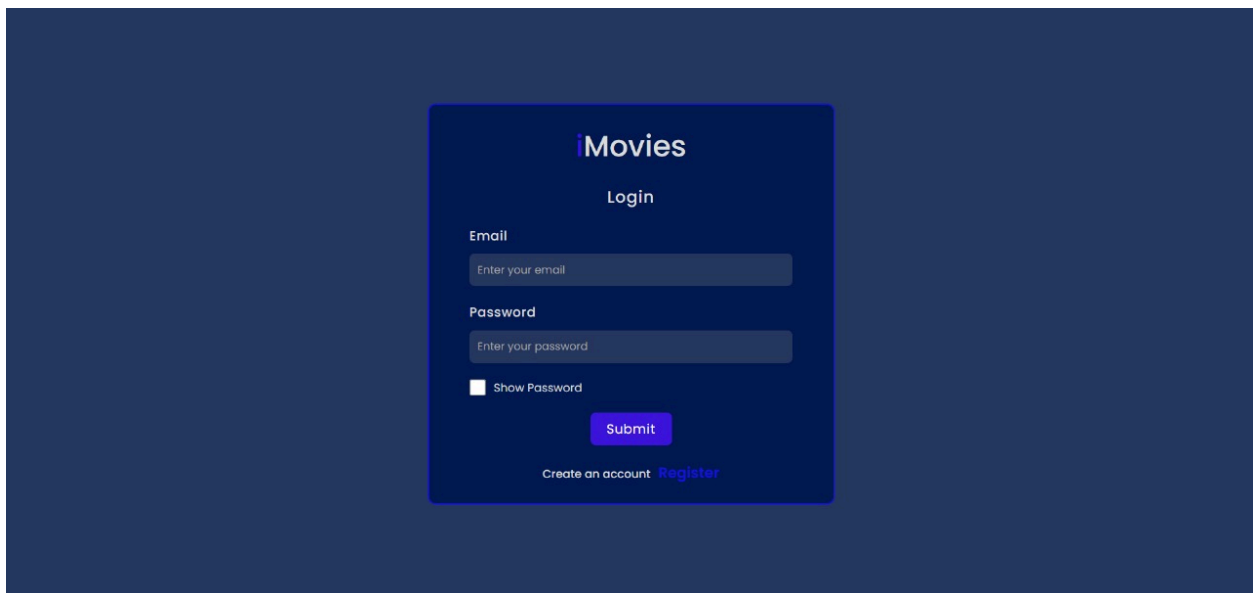
☐ Show Password

Submit

Already have an account? [Login](#)

Register as an [Admin](#)

iMovies Registration Page



The screenshot shows a login form for 'iMovies' on a dark blue background. The form is centered and contains the following elements: a title 'iMovies' with a logo, a subtitle 'Login', two input fields for 'Email' and 'Password', a 'Show Password' checkbox, a 'Submit' button, and a 'Register' link.

iMovies

Login

Email
Enter your email

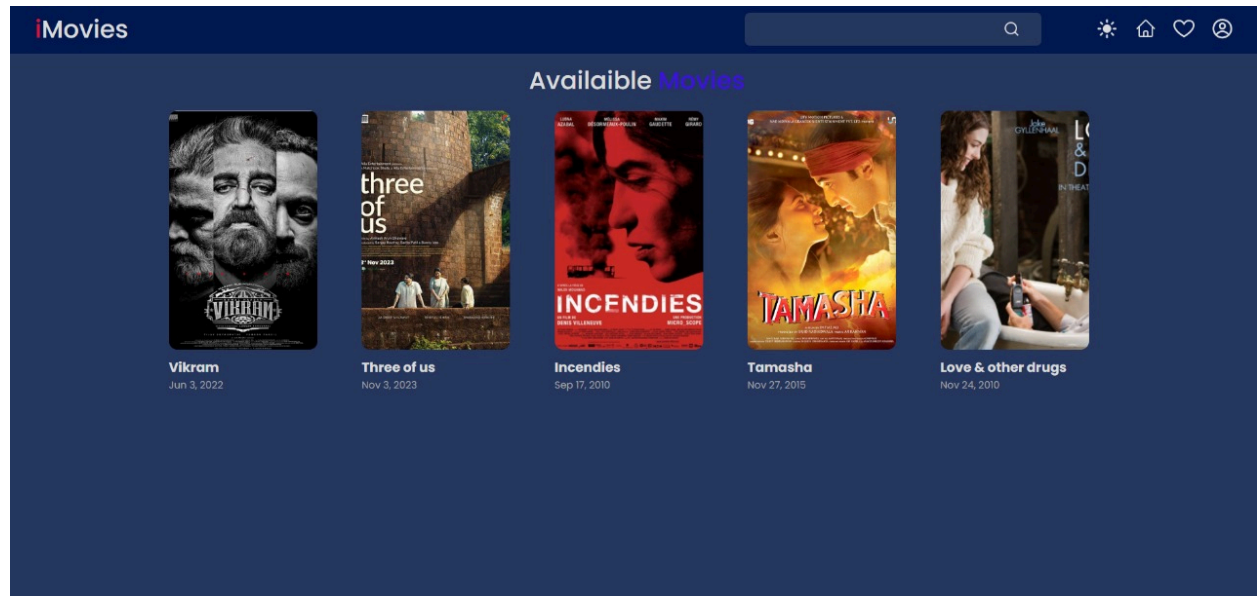
Password
Enter your password

☐ Show Password

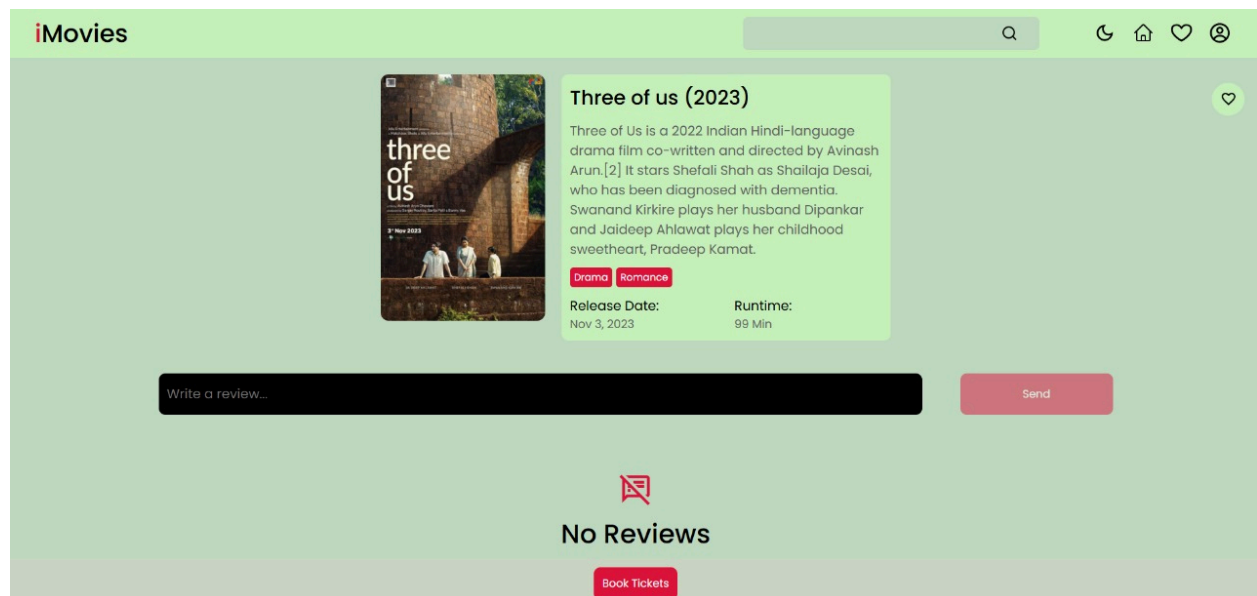
Submit

Create an account [Register](#)

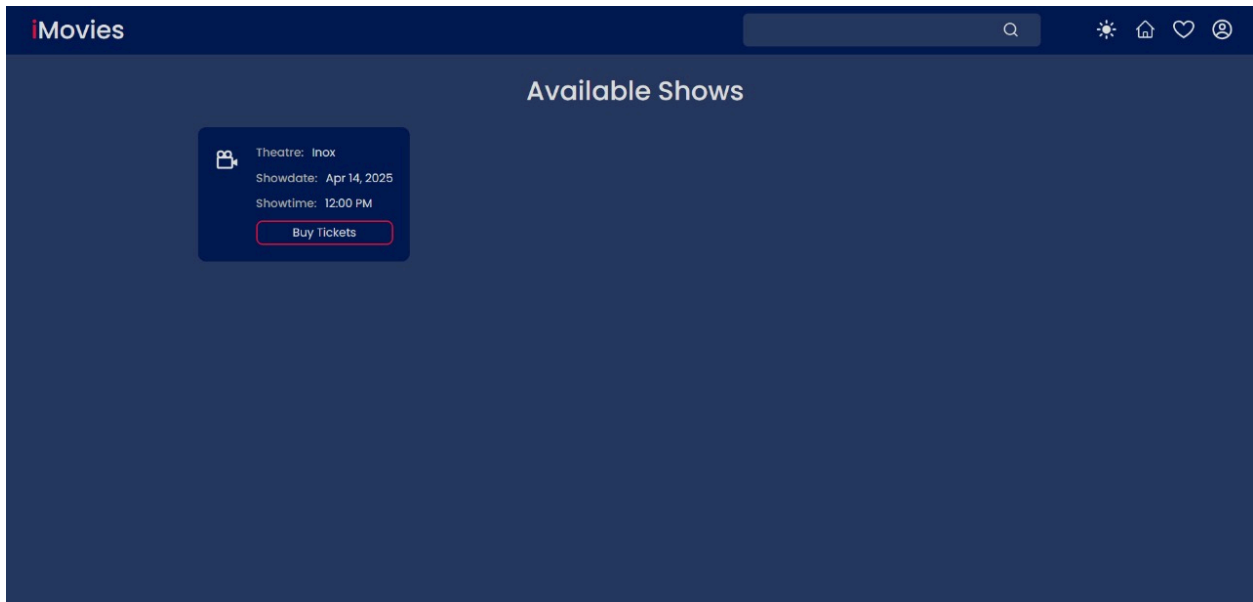
iMovies Login Page



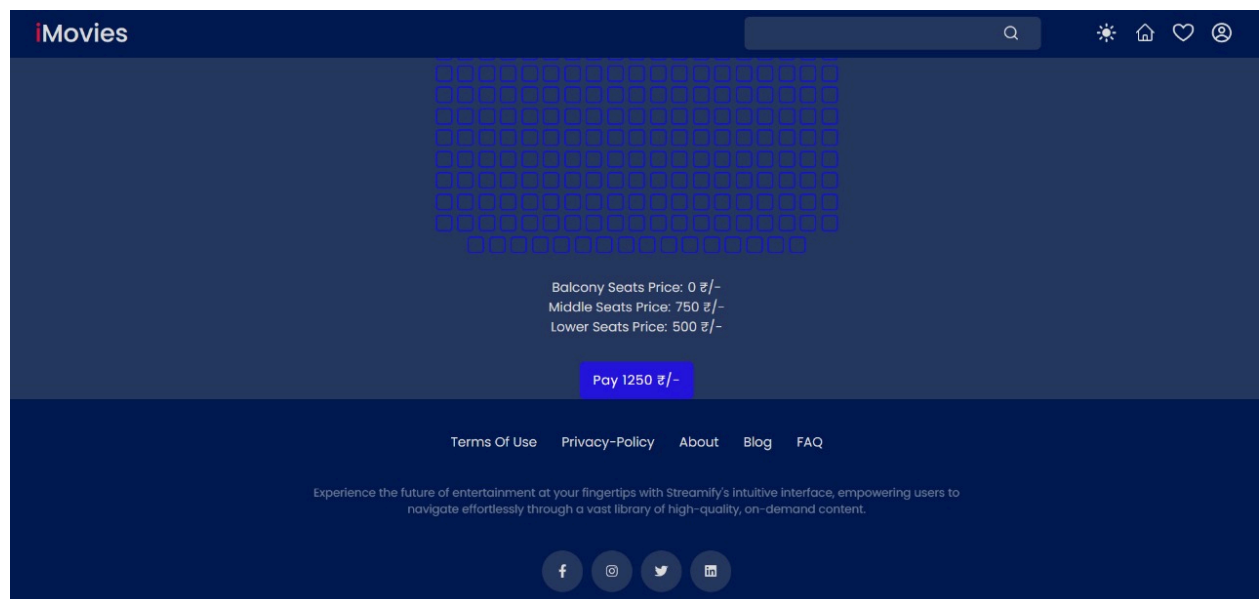
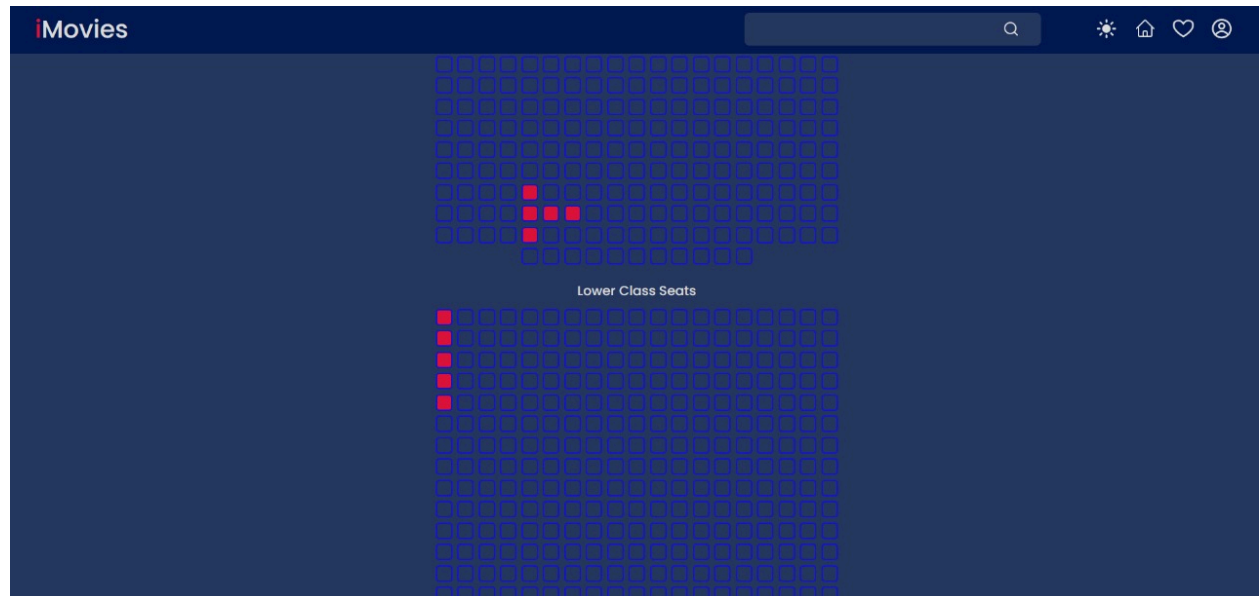
Home Page



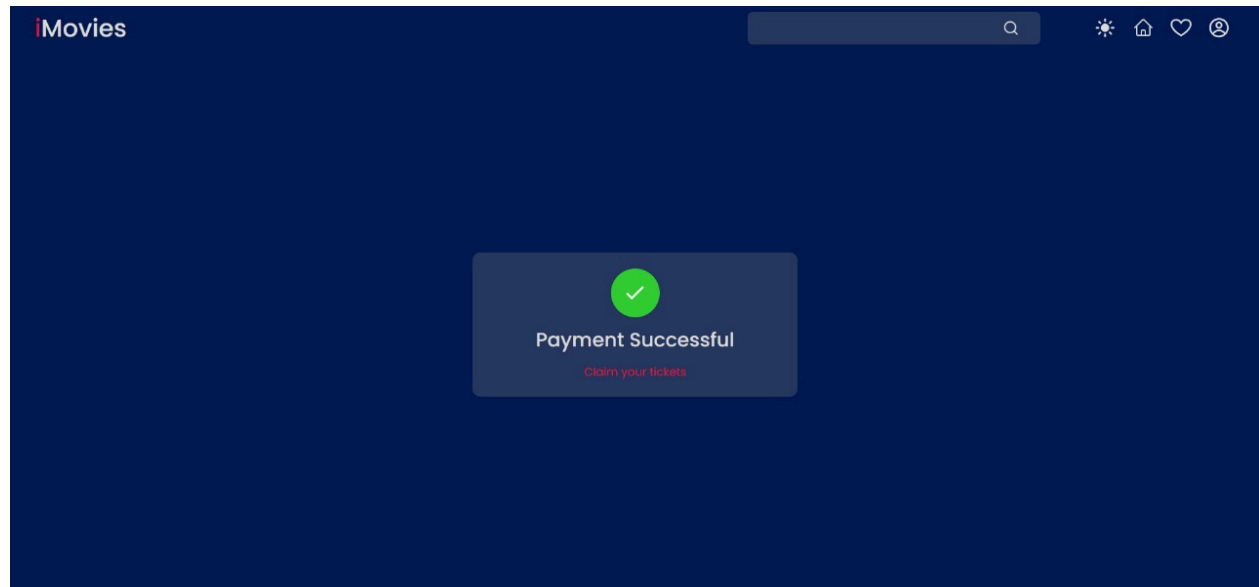
Movie Detailed Page



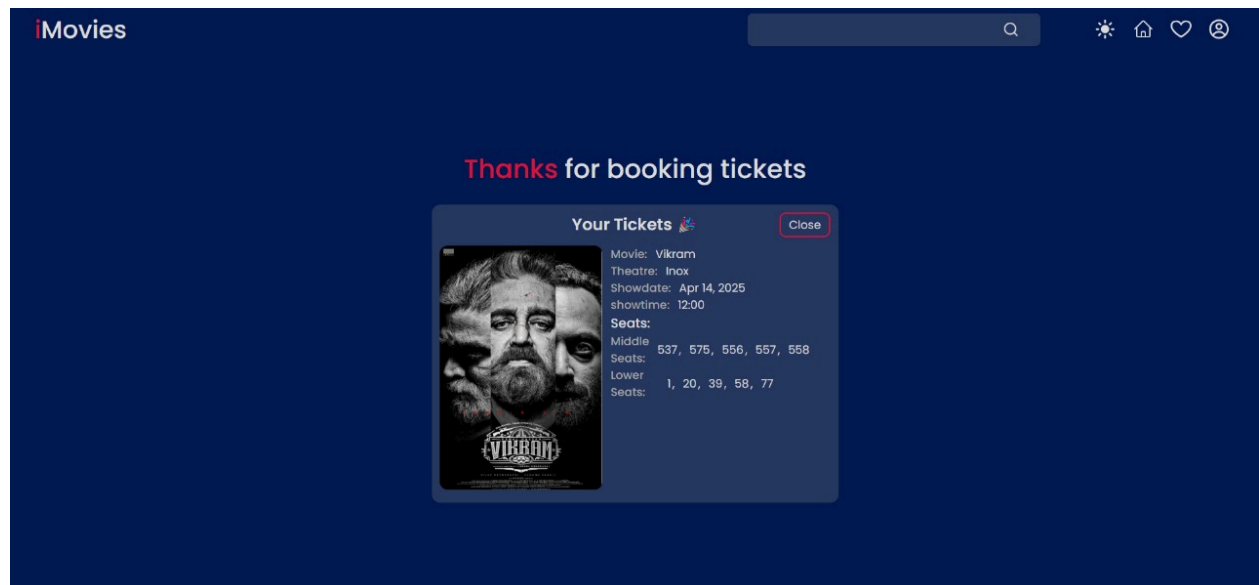
Available Show showing Page



Seat Selection Page



Successfully Payment Done



Booking Confirmation Page

12. Known Issues

- No payment integration yet
 - Admin panel UI is basic
 - Email confirmation not implemented
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13. Future Enhancements

- Integrate payment gateway (e.g.Razorpay)
- Add user reviews and rating system
- Enable email booking confirmations
- Create mobile app version (React Native)
- Improve admin panel UI and access control